Switches that decide

IoT · Level 2 · Lesson 2



Today you will

- Explain AND, OR and NOT with switches and lights.
- Fill in a truth table for a simple rule.
- Design the logic for a real device, like a fan or an alarm.

AND, OR and NOT

- Series makes AND
- Parallel makes OR
- NOT flips its input
- A truth table checks every case

Truth table detective

1. Pick three devices that decide
2. Write each rule in words
3. Turn it into AND, OR or NOT
4. Make the truth table
5. Find the silly row and fix it

No kit needed. If you build with batteries: batteries only, press the paper clips onto foil strips to make switches, and never join the battery's + and - with foil alone.

Let’s discuss

1. Which device's rule was hardest to write, and why?
2. Where would a mistake in a rule waste money or be unsafe?
3. Why is it worth writing the truth table before you build?

Check yourself

- Write the truth table for A AND NOT B.